WK 7 / 12

Why systems beat gut feel

One rule, read the same way over every company every night — and the evening we find out what a rule is, what it costs to follow one, and what it cannot know

We grade decisions, not outcomes.

Education, not advice.

0:00 — HARVEST

Your 120 words

Two read aloud

Every sentence with its fact tag. There is usually one sentence without one. Find it together.

The machine did it to itself

24 Oct: the AI wrote The Andersons' revenue as 11.2 billion. The sheet said 11.3. The toolkit flagged it. Your exercise, on a live page, before any room existed.

Tonight

From reading to deciding. The first decision is who decides: you, on the night — or a rule, written down before.

0:10 — BEFORE ANYONE SEES A CHART

JOURNAL FIRST

Predict first

On your slip: three companies from the universe you believe are going up — and three you believe are going down.

Name. Date. Fold it.

Your gut, on the record, before the rule speaks. They come back at 0:45.

0:15 — CARVER'S FIRST CHAPTER

The flawed human brain

We trade too much

66,000 households, 1991–96. The fifth that traded most earned 11.4% a year. The market made 17.9%. Overconfidence, measured in commissions.

Barber & Odean, 2000

We sell winners, keep losers

The same kind of records (10,000 accounts, 1987–93): about 50% more likely to sell a share showing a gain than one showing a loss. Loss aversion with a brokerage account.

Odean, 1998

A rule has none of this

It cannot be overconfident, feel a loss, or read the news. It does the same thing on a bad night as on a good one. What it CAN be is wrong — openly, measurably, testably.

Not that rules are right. That rules can be checked, and gut feel cannot.

Systematic Trading ch. 1; Barber & Odean, J. Finance 55(2); Odean, J. Finance 53(5)

The arithmetic, in full

Step	What it is	The number
λ	the weight on today: 2 ÷ (span + 1)	16 days: 2/17 = 0.117647 · 64 days: 2/65 = 0.030769
EWMA	λ × today's close + (1 − λ) × yesterday's average	start both at the first close
σ	EW standard deviation of daily price changes	span 35 — Carver's default
forecast	$3.75 \times (EWMA_{16} - EWMA_{64}) \div \sigma$	capped at ±20; 3.75 is Carver's scalar
state	forecast above zero → LONG; otherwise FLAT	long-only: no shorts, no leverage (ours)
instrument risk	σ of daily % returns × 16 (√256 days)	a year's volatility, as a %

The fast average moves first. A crossover is the fast one overtaking the slow one — which only happens after a trend has been going a while. Late on purpose: being late is what filters the noise.

Leveraged Trading — the Starter System; Systematic Trading ch. 7; pysystemtrade (GPL)

BY HAND

Three closes: 10, 11, 12

Day	Close	$EWMA_{16}$ ($\lambda = 2/17$)	$EWMA_{64}$ ($\lambda = 2/65$)
1	10	10 (start)	10 (start)
2	11	(2 × 11 + 15 × 10) ÷ 17 = 10.117647	(2 × 11 + 63 × 10) ÷ 65 = 10.030769
3	12	(2 × 12 + 15 × 10.117647) ÷ 17 = 10.339100	(2 × 12 + 63 × 10.030769) ÷ 65 = 10.091361

After day 3 the fast average is already 0.25 above the slow one. Use the fractions: a four-place λ moves the last digit. The rule is not yet trusted — it needs 20 changes for σ and 128 closes before it calls a state — but you can see which way it leans, and you can check every digit.

0:45 — LIVE

The rule, over the whole universe, tonight

`[ course URL ]/signals`

`[ course URL ]/prices/AAPL`

Read the tally first: how many long, how many flat, how many warming up. Then one chart: the blue 16-day line, the red dashed 64-day line, the strip underneath — green where the rule was long, grey where it was flat. Hover the strip for the dates.

Then the slips. Keep the tally on the board: agreed with the rule · disagreed. Expect about half. So is the rule, on any one day. It will do the same thing tomorrow; you will not.

Education, not advice.

1:00 — THE NUMBER EVERYTHING HANGS OFF

Instrument risk

What it is

The annualised volatility of the share: the EW standard deviation of daily returns × 16. A share at 30% moves about 30% a year, two years in three — up or down.

Read it off the page

Household names typically 20–30%. Small caps 40% and up. Rank the universe by it in your head; the rule already has.

Next week

That one number becomes a position size — a volatile share gets a smaller holding — and a stop: half the annual risk below the high. Everything in Phase 3 hangs off it.

Reproduce the machine

The data

[course URL]/api/prices/AAPL?days=1900&csv=1 — the whole five years (a one-year sheet still remembers its start in the slow average). Open it in a spreadsheet; use column F, adj_close.

The columns

$EWMA_{16}$ · $EWMA_{64}$ · price change · variance ($\lambda \times change^2 + (1 - \lambda) \times$ yesterday's) · $\sigma = \sqrt{variance}$ · forecast · state.

The check

The last row must match /api/signals/AAPL → latest: the averages and σ to six decimals, the forecast to two. It will. A rule is an answer you can reproduce; an oracle is one you can't.

The syllabus's exit check for this week, done by the room instead of the tutor.

SAID OUT LOUD

What this is not

Not advice

A long is not “buy”; a flat is not “sell”. The rule has read 64 days of closes and divided by the noise. It knows nothing else.

Not a backtest

The strip shows when the rule was long. It shows no returns, no costs, no stops. Trend rules like this one have made modest money across many markets over a century of data, with long, deep drawdowns — and can lose on any one share in any one year.

Not finished

No sizing, no stops, no tickets — week 8. The small-cap tier is warming up until 128 closes are stored, and is not split-adjusted: a split would show as a false flip (on the register).

Hurst, Ooi & Pedersen (2017), J. Portfolio Management 44(1)

1:25 — HOMEWORK

Your one instrument

1 · The rule in your words

Five lines, no symbols, as you would explain it to a friend. Bring it.

2 · Pick your one instrument

The company you will follow with this rule for the rest of the course. Not the one you like: the one whose books, chart and flips you have read. Why, in the journal.

3 · Read

The Starter System's opening-rule and instrument-risk chapters (course copy). Optional, the science: Hurst, Ooi & Pedersen, “A Century of Evidence on Trend-Following Investing” — free on AQR's site.

“We grade decisions, not outcomes.”

GO DEEPER — FREE, FOREVER, OPTIONAL

The science, and the code

Everything the rule rests on is public and free to read:

PAPER 2000 Barber & Odean, “Trading Is Hazardous to Your Wealth” — what trading on feel costs, measured

PAPER 2017 Hurst, Ooi & Pedersen, “A Century of Evidence on Trend-Following Investing” — 1880–2016, 67 markets

CODE pysystemtrade — Carver's own open-source system (GitHub, GPL) — the constants the toolkit uses, in his hand

CORE UNIT 11 Asset markets and bubbles (The Economy 2.0) — stories, herds and the price of a share

github.com/robcarver17/pysystemtrade · books.core-econ.org/the-economy

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